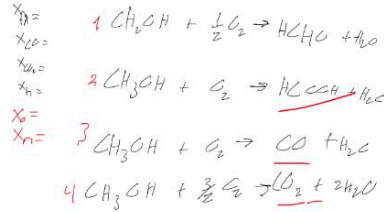


D.C.F

9 3
4 2
7



equations:

$$M_{\text{out}} = M_{\text{in}}(1 - 0.9) = 60 \frac{\text{kmol}}{\text{hr}}$$

$$M_{\text{out}} = 100(1 - 0.95) = 4.75 \frac{\text{kmol}}{\text{hr}}$$

$$\begin{aligned} \text{CH}_3\text{OH}: M_{\text{out}} &= M_{\text{in}} - \left[\frac{6}{1} - \frac{6}{2} - \frac{6}{3} - \frac{6}{4} \right] \\ \text{O}_2: M_{\text{out}} &= M_{\text{in}} - \left[\frac{1}{2} \frac{6}{1} - \frac{6}{2} - \frac{6}{3} - \frac{3}{2} \frac{6}{4} \right] \end{aligned}$$

$$\text{H}_2\text{O}: M_{\text{out}} = \frac{6}{1} + \frac{6}{2} + \frac{6}{3} + \frac{6}{4} = 4.75 \text{ kmol}$$

$$\text{HCHO}: M_{\text{out}} = \frac{6}{1} = 4.75 \text{ kmol}$$

$$\text{HCOOH}: M_{\text{out}} = \frac{6}{2} = 3$$

$$\text{CO}: M_{\text{out}} = \frac{6}{3} = 2$$

$$\text{CO}_2: M_{\text{out}} = \frac{6}{4} = 1.5$$

$$[\text{CH}_3\text{OH}] \quad 40 = \frac{6}{1} + \frac{6}{2} + \frac{6}{3} + \frac{6}{4}$$

$$[\text{O}_2] \quad 20 = \frac{1}{2} \frac{6}{1} + \frac{6}{2} + \frac{6}{3} + \frac{3}{2} \frac{6}{4}$$

$$20 = \frac{1}{2} \frac{6}{1} - \frac{1}{2} \frac{6}{4}$$

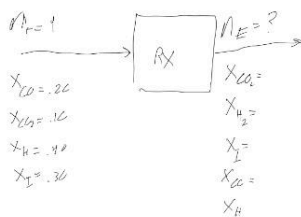
Assume all side reactions are zero

$$20 \cdot 2 = \frac{6}{1} = 40$$

No reactor outlet compositions, -5 pts

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2)



a)

DOF:

$$(4 + 4) - (5 + 5 + 5 + 5) = 0$$

$$(4 + 1) - (1 + 3 + 1 + 5) = 0$$

$$0 - 0 = 0, \text{ well specified}$$

b)

species:

$$\text{CO: } \dot{N}_{\text{CO, out}} = \dot{N}_{\text{CO, in}} - \dot{N}_{\text{CO, cons}} = 0.2 - 0.1098 = 0.0902$$

$$\text{CO}_2: \dot{N}_{\text{CO}_2, \text{ out}} = \dot{N}_{\text{CO}_2, \text{ in}} + \dot{N}_{\text{CO}_2, \text{ prod}} = 0.3 + 0.1098 = 0.4098$$

$$\text{H}_2: \dot{N}_{\text{H}_2, \text{ out}} = \dot{N}_{\text{H}_2, \text{ in}} - \dot{N}_{\text{H}_2, \text{ cons}} = 0.4 - 0.1098 = 0.2902$$

$$\text{H}_2\text{O: } \dot{N}_{\text{H}_2\text{O, out}} = \dot{N}_{\text{H}_2\text{O, in}} + \dot{N}_{\text{H}_2\text{O, prod}} = 0.1 + 0.1098 = 0.2098$$

$$\text{I: } \dot{N}_{\text{I, out}} = \dot{N}_{\text{I, in}} = 0.30$$

total moles? -5 pts

$$\frac{X_{\text{CO}_2} X_{\text{H}_2}}{X_{\text{CO}} X_{\text{H}_2\text{O}}} = K_p = 0.2$$

$$\frac{(0.10 + y)(y)}{(0.20 - y)(0.4 - y)} = 0.2 \quad (400/103)$$

$$= 0.0247 \quad (3.58)$$

$$0.1y + y^2 = 0.8472(0.08 - 0.6y + y^2)$$

$$0.1128y^2 + 0.63y - 0.071 = 0$$

$$y = 0.1098 = \dot{N}_{\text{H}_2, \text{ cons}}$$

c)

The reader might not have reached equilibrium yet.

d)

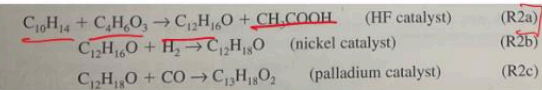
Temp is changing						
T	xCO	xCO2	xH2O	K	xH2	
1023	0.3	0.3	0.4	1.256923	0.502769	
1123	0.3	0.3	0.4	0.885806	0.354323	
1223	0.3	0.3	0.4	0.661032	0.264413	

Mole concentrations changing						
T	xCO	xCO2	xH2O	K	xH2	
1123	0.5	0.5	1.00E-05	0.885806	8.86E-06	
1123	0.4	0.3	0.3	0.885806	0.354323	
1123	0.2	0.2	0.6	0.885806	0.531485	

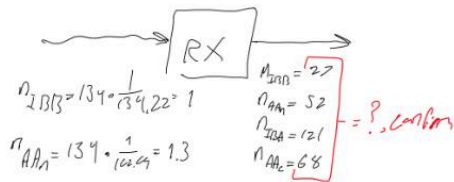
if we decreased temperature and increased the mole percent of CO and H₂O, we can maximize the H₂ content

20/25

P4.10 Your company is interested in building a plant to make ibuprofen, using the new catalytic reaction scheme described in Example 4.1. Your job is to measure the reaction rate with different catalysts, as a first step in designing a full-scale reactor. In one experiment, you mix 134 g isobutylbenzene (IBB, $C_{10}H_{14}$) with 134 g acetic anhydride (AAn, $C_4H_6O_3$) in a laboratory-scale batch reactor, adjust the temperature, add some catalyst, and wait 1 h. At the end of the hour you stop the reaction, collect all the material in the pot, and send it for chemical analysis. The chemist reports back that the pot contains 27 g IBB, 52 g acetic anhydride, 121 g isobuty-lacetophenone (IBA, $C_{12}H_{16}O$), and 68 g acetic acid (AAc, CH_3COOH). When you see these results you tell the chemist that the analysis is wrong. He protests. Who's right?



DOF



$$(6 + 1) - (1 + 1 + 1 + 4) = 1, \text{ under}$$

bas. S: $n_{IBB, \text{out}} = 27g \cdot \left(\frac{1}{134.22}\right) = 0.2 \text{ mol IBB}$

$$U = i_n - o_n + f_{gen} - c_n$$

IBB: $n_{IBB, \text{out}} = n_{IBB, \text{in}} - \sum_1 \rightarrow \sum_1 = n_{IBB, \text{in}} - n_{IBB, \text{out}} = 0.8$

AAn: $n_{AAn, \text{out}} = n_{AAn, \text{in}} - \sum_1 \rightarrow n_{AAn, \text{out}} = (1.3 - 0.8) \cdot \frac{102.09}{1} = 51g$

IBA: $n_{IBA, \text{out}} = \sum_1 \rightarrow n_{IBA, \text{out}} = 0.8 \cdot (176.26) = 141g$

AAc: $n_{AAc, \text{out}} = \sum_1 \rightarrow n_{AAc, \text{out}} = 0.8 \cdot (60.052) = 48g$

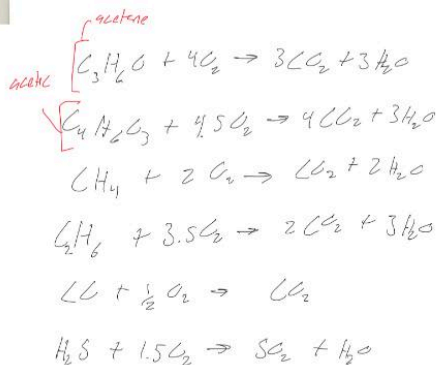
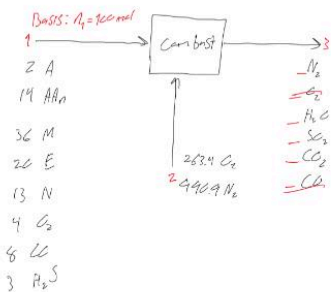
We are correct, the analysis is wrong



25/25

P4.19 Waste streams from various processes in a complex chemical facility are combined and burned for heat, because the cost of purifying and recycling the components is prohibitive. On a typical day, the combined waste stream contains 2 mol% acetone (CH_3CO), 14 mol% acetic anhydride ($\text{CH}_3\text{CO}_2\text{O}$), 36 mol% methane (CH_4), 20 mol% ethane (C_2H_6), 13 mol% N_2 , 4 mol% O_2 , 8 mol% CO , and 3 mol% H_2S . The waste stream is fed to a furnace along with 20% excess air and combusted. (For combustion reactions, excess air is calculated on the assumption that the only products of combustion are CO_2 , H_2O , N_2 , and SO_2 .) Analysis of the exit flue gas gives a CO_2/CO mass ratio of 4:1. The only other components in the flue gas are N_2 , O_2 , H_2O , and SO_2 . Calculate the composition of the exit flue gas. Assume air is 21 mol% O_2 and 79 mol% N_2 .

based on EX 4.7
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$$\dot{V}_{\text{O}_2, \text{Air}} = (2 \cdot 4 + 14 \cdot 4.5 + 36 \cdot 2 + 20 \cdot 3.5 + 8 \cdot \frac{1}{2} + 3 \cdot 1.5 + 4) (1 + 0.2)$$

$$\dot{V}_{\text{O}_2, \text{Air}} = 261 \text{ mol}$$

$$\dot{V}_{\text{N}_2, \text{Air}} = 261 \cdot \frac{0.79}{0.21} = 981.4 \text{ mol}$$

Atomic balance ($\text{N}_2, \text{O}_2, \text{H}_2, \text{S}, \text{C}$)

N: $\dot{n}_{\text{N}_2, \text{out}} = \dot{n}_{\text{N}_2, \text{in}} + \dot{n}_{\text{N}_2, \text{Air}} = 450 + 981.4 \rightarrow \dot{n}_{\text{N}_2, \text{out}} = 1431.4 \text{ mol}$

S: $\dot{n}_{\text{H}_2\text{S}, \text{in}} = \dot{n}_{\text{SO}_2, \text{out}} \rightarrow \dot{n}_{\text{SO}_2, \text{out}} = 3 \text{ mol}$

H: $6\dot{n}_{\text{A}, \text{in}} + 6\dot{n}_{\text{AA}, \text{in}} + 4\dot{n}_{\text{M}, \text{in}} + 6\dot{n}_{\text{E}, \text{in}} + 2\dot{n}_{\text{H}_2\text{S}, \text{in}} = 2\dot{n}_{\text{H}_2\text{O}}$

$$6 \cdot 2 + 6 \cdot 14 + 4 \cdot 36 + 6 \cdot 20 + 2 \cdot 3 = 2\dot{n}_{\text{H}_2\text{O}} \rightarrow \dot{n}_{\text{H}_2\text{O}} = 143 \text{ mol}$$

C: $3\dot{n}_{\text{A}, \text{in}} + 4\dot{n}_{\text{AA}, \text{in}} + \dot{n}_{\text{M}, \text{in}} + 2\dot{n}_{\text{E}, \text{in}} + \dot{n}_{\text{CO}, \text{in}} = \dot{n}_{\text{CO}_2, \text{out}} + \dot{n}_{\text{CO}, \text{out}}$

$$3 \cdot 2 + 4 \cdot 14 + 36 + 2 \cdot 20 + 8 = 146 = \dot{n}_{\text{CO}_2} + \dot{n}_{\text{CO}}$$

$$\text{CO}_2 = 4\text{CO}$$

$$\frac{\dot{n}_{\text{CO}_2}}{\dot{n}_{\text{CO}}} = \frac{4}{1} \quad \frac{\dot{n}_{\text{CO}_2}}{\dot{n}_{\text{CO}}} = \frac{4 \cdot 260}{1 \cdot 114} = 2.55$$

$$2.55 \text{ CO}_2 = \text{CO}$$

$$146 = 3.55 \text{ CO}_2$$

$$\dot{n}_{\text{CO}_2, \text{out}} = 41.1$$

$$\dot{n}_{\text{CO}, \text{out}} = 164.9$$

O: $\dot{n}_{\text{O}_2, \text{in}} + 3 \cdot \dot{n}_{\text{AA}, \text{in}} + 2 \cdot \dot{n}_{\text{CO}_2, \text{in}} + \dot{n}_{\text{CO}, \text{in}} + 2\dot{n}_{\text{O}_2, \text{Air}} = 2\dot{n}_{\text{H}_2\text{O}, \text{out}} + \dot{n}_{\text{H}_2\text{O}} + 2 \cdot \dot{n}_{\text{SO}_2} + 2\dot{n}_{\text{CO}_2, \text{out}} + \dot{n}_{\text{CO}}$

$$2\dot{n}_{\text{O}_2, \text{Air}} = 2 \cdot 261 + \dot{n}_{\text{CO}_2, \text{out}} + 2\dot{n}_{\text{CO}_2, \text{in}} + 129$$

$$\dot{n}_{\text{O}_2, \text{Air}} = \frac{2 \cdot 146 + 129}{2} = 20 \text{ mol} \cdot (1 + 0.2) = 252 \text{ mol}$$

$$\dot{n}_{\text{N}_2, \text{Air}} = 210.5 \text{ mol} \cdot \frac{79}{21} = 792 \text{ mol} \cdot (1 + 0.2) = 950 \text{ mol}$$

$$\dot{n}_{\text{CO}_2, \text{out}} = 62.65 \text{ mol}$$

ex. -
The flute composition

$N_2 = 963 \text{ mol}$ $SO_2 = 3 \text{ mol}$ $H_2O = 183 \text{ mol}$ $CO_2 = 41.1 \text{ mol}$

$CO = 104.4$ $O_2 = 62.65$

$TOT = 1352.7 \text{ mol}$

$X_{N_2} = 70.9\% \checkmark$

$X_{SO_2} = 0.2\% \checkmark$

$X_{H_2O} = 13.5\% \checkmark$

$X_{CO_2} = 3\% \checkmark$

$X_{CO} = 7.7\% \checkmark$

$X_{O_2} = 4.6\% \checkmark$

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